Stable, Threshold-Aligned AMR Policy Signals from Routine Surveillance

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Abstract

Threshold-based AMR guideline triggers are vulnerable to year-to-year noise and uneven sample sizes, making provincial decisions brittle. We present a general, uncertainty-aware decision rule that converts routine antibiograms into stable, threshold-aligned policy signals. The method fits a hierarchical Empirical-Bayes Beta–Binomial model to borrow strength from national totals, reports Jeffreys 95% credible intervals, and, for each organism–drug–specimen and year, computes the exceedance probability $\Pr(\theta > \tau)$ relative to syndrome-specific concern thresholds τ. To suppress one-off spikes, we add stability logic: a persistence rule requiring $\Pr(\theta > \tau) \geq 0.80$ for $K = 2$ consecutive years plus a 3-year positive-slope check on the EB estimate. Policy impact is expressed as excess failures per 1,000 tests, calculated as $1000 \cdot \max(\hat{\theta} - \tau, 0)$ using the EB mean. The pipeline outputs decision-ready tables—EB summaries (means, intervals, exceedance), stability flags, an actionable change list ranked by impact/1,000 and confidence, and a simple mismatch index—and runs end-to-end on standard CSVs without MCMC. Applied to data from the National Institute for Communicable Diseases (NICD) Antimicrobial Resistance Surveillance Dashboard (NHLS public-sector data, 2021–2024), stratified by province, the approach generates stable, threshold-aligned resistance signals and a concise, rank-ordered list of candidate guideline revisions suitable for scheduled antimicrobial stewardship review. The method is implemented in an open, reproducible analytical notebook.

Significance.

- Turns routine lab data into clear, *stable* signals for when guidelines likely need changing.
- Reduces noise from small samples by borrowing national information and requiring persistence across years.
- Quantifies the chance of being over the threshold and converts it into expected extra failures per 1,000 tests.
- Produces a ranked, ready-to-act list so provinces can target updates and monitoring efficiently.

Keywords: antimicrobial resistance; empirical Bayes; policy thresholds; policy

1 Introduction

Antimicrobial stewardship programmes commonly use fixed resistance thresholds to decide when first-line recommendations should change [1]. In practice, however, routine surveillance data are noisy and unevenly sampled across provinces: some organism–drug–specimen cells include thousands of isolates, while others have only a few dozen [2]. Raw annual resistance percentages fluctuate substantially at small sample sizes, and simple rolling averages can obscure genuine changes while still allowing transient spikes to trigger premature policy changes [3–5]. This creates brittle decision-making, with a risk of false "change-guideline" alerts in low-n settings and missed signals where real resistance drifts are emerging [6].

Conventional summaries—raw proportions with confidence intervals or simple smoothing—do not directly answer the stewardship question of interest: *what is the probability that true resistance in this province, this year, exceeds the clinical threshold?* [7] Nor do they guard explicitly against transient spikes or provide an interpretable measure of policy impact. A method that (i) aligns to the clinical threshold, (ii) accounts for small-n uncertainty, and (iii) incorporates temporal persistence is therefore needed to transform routine antibiograms into reliable, repeatable stewardship signals [8].

To address this, we introduce a general, uncertainty-aware decision rule that converts provincial antibiograms into threshold-aligned policy signals. The method combines: (1)

hierarchical Empirical-Bayes Beta–Binomial estimation to borrow statistical strength from national totals and stabilise small-n cells [9–11]; (2) the exceedance probability $\Pr(\theta > \tau)$ relative to syndrome-specific thresholds τ drawn from current treatment guidance [12]; (3) a persistence rule requiring $\Pr(\theta > \tau) \geq 0.80$ for $K = 2$ consecutive years, supported by a three-year positive-slope check to suppress one-year spikes [13, 14]; and (4) an interpretable impact metric, excess failures per 1,000 tests, calculated as $1000 \cdot \max(\hat{\theta} - \tau, 0)$ using the Empirical-Bayes mean. All quantities are closed form and computationally efficient using standard surveillance tables. We apply this framework to national public-sector antimicrobial resistance data from the NICD/NHLS surveillance system (2021–2024), stratified by province and common organism–drug–specimen combinations. These data represent South Africa's primary public-sector AMR surveillance dataset and provide a realistic testbed for threshold-aligned, uncertainty-aware decision metrics [8].

2 Methods

2.1 Study design and data source

We conducted a secondary analysis of routine public-sector antimicrobial resistance (AMR) surveillance data from the National Institute for Communicable Diseases / National Health Laboratory Service (NICD/NHLS), stratified by province, organism, antibiotic, specimen type, and year (2021–2024) [15, 16]. For the main results we focus on **blood culture** exports available via the public dashboard [17]. For each province–pair–specimen–year cell we extracted the total tested (n) and the percentage resistant ($\%R$). When only $\%R$ was given, resistant counts were back-calculated as

$$r = \mathrm{round}\big(n \times (\%R/100)\big).$$

The working schema is: `province`, `organism`, `antibiotic`, `specimen`, `year`, n, r (and optional `sector`).

2.2 Pre-processing and inclusion

We restricted the analysis to `BLOOD` specimens and the `combined` sector. Province name variants were harmonised to a standard set. Any duplicate rows for a given province–pair–specimen–year were resolved by keeping the last entry. Rows with missing or zero counts are handled defensively in subsequent estimators, consistent with standard quality control in public-health analytics [5, 14].

2.3 Threshold policy

Concern thresholds (τ) are **syndrome-level** policy targets learned from the most recent year using a simple grid search over $\tau \in [0.10, 0.40]$ to align with target failure tolerances [7]. **Specifically, we select the τ that minimizes the absolute difference between the target and the mean shrunken resistance ($\hat{\theta}$) among all provinces flagged as exceeding that τ.** For this analysis, bug–drug pairs were mapped to syndromes as follows: *E. coli*–ciprofloxacin → UTI; *S. aureus*–cefoxitin → SSTI; and *S. pneumoniae*–penicillin → CAP. The learned values were then mapped onto each record as `tau_used`.

2.4 Empirical–Bayes estimation (national Bayes-Laplace prior)

Within each year and organism–drug pair, national totals were computed as $R_{\text{nat}} = \sum r$ and $S_{\text{nat}} = \sum (n - r)$. We formed a **Bayes-Laplace (uniform) prior** by adding 1.0 pseudo-counts to each side:

$$\alpha_0 = R_{\text{nat}} + 1.0, \qquad \beta_0 = S_{\text{nat}} + 1.0.$$

For province p with r_p resistant of n_p tested, the EB posterior parameters are

$$a_p = \alpha_0 + r_p, \qquad b_p = \beta_0 + (n_p - r_p),$$

with EB mean $\hat{\theta}_p = \frac{a_p}{a_p + b_p}$ [18–20].

2.5 Uncertainty and exceedance probability

For display of interval uncertainty around the **raw provincial proportion**, we report a **95% Beta-posterior credible interval** using $a = r_p + 0.5$, $b = (n_p - r_p) + 0.5$. For **decision-making**, we compute the **exceedance probability** that true resistance exceeds the concern threshold from the **EB posterior**:

$$\Pr(\theta_p > \tau) = 1 - F_{\text{Beta}}(\tau;\, a_p,\, b_p),$$

where F_{Beta} is the Beta CDF with parameters (a_p, b_p) defined above [21].

2.6 Stability rule (persistence OR slope)

To avoid spurious one-year alerts, a province–pair–year is promoted to `stable_alert` if **either** condition holds:

1. **Persistence:** $\Pr(\theta > \tau) \geq 0.80$ for $K = 2$ consecutive years (run-length rule); *or*
2. **Positive trend:** the 3-year slope from a simple linear fit of the **exceedance probability** ($\Pr(\theta > \tau)$) is ≥ 0.05.

This approach aligns with classical empirical-Bayes stability formulations and recent work on sequential decision rules in surveillance [22, 23].

2.7 Policy impact metric

Among alerts, we prioritise by **excess failures per 1,000 tests**,

$$\text{Impact}/1{,}000 = 1000 \cdot \max\big(\hat{\theta}_p - \tau,\, 0\big),$$

an interpretable burden metric for empiric-therapy review [8].

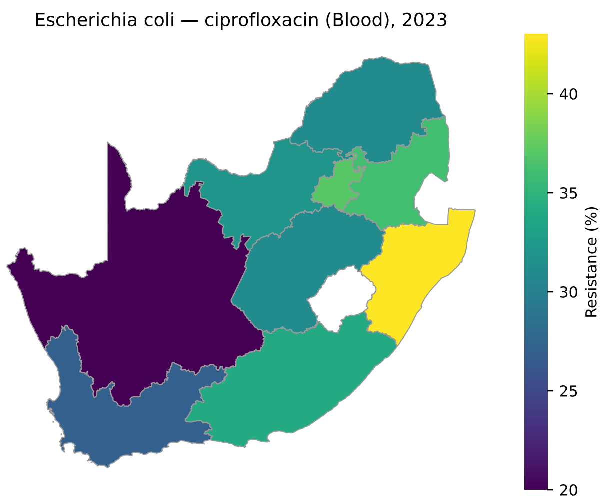

Figure 1: **Provincial overview (2023).** *E. coli*–ciprofloxacin (blood culture), 2023. Shading reflects provincial resistance, illustrating spatial heterogeneity and the need to account for uneven sample sizes in decision rules.

2.8 Software and reproducibility

All analyses run in a containerised Python environment (Docker/Compose spec included) [24]. Calculations are closed-form (no MCMC) and reproduce from CSV inputs. An anonymised archive containing the notebook, outputs, and container spec accompanies the submission.

3 Results

3.1 Provincial overview (2023)

Figure 1 summarises provincial resistance for *Escherichia coli*–ciprofloxacin (blood culture) in 2023. The geographic pattern is heterogeneous, with visibly higher resistance in several provinces and notable differences in sampling intensity. This motivates a threshold-aligned, uncertainty-aware approach rather than relying on raw annual percentages alone, particularly for provinces with small n.

3.2 Stable, threshold-aligned impact (2024)

For 2024, all nine provinces exceeded the concern threshold $\tau = 10\%$ by large margins when estimated with Empirical–Bayes (EB) shrinkage. The EB point estimates range from **32.6%** (Gauteng) to **35.7%** (KwaZulu–Natal), implying absolute exceedances $\Delta = \hat{\theta} - \tau$ between **22.6** and **25.7** percentage points and corresponding impacts of **226.2** to **256.6** excess failures per 1,000 tests. Given these deltas, exceedance probabilities $\Pr(\theta > \tau)$ are near-certain, and ranking is driven primarily by the magnitude of Δ and the stability logic (persistence).

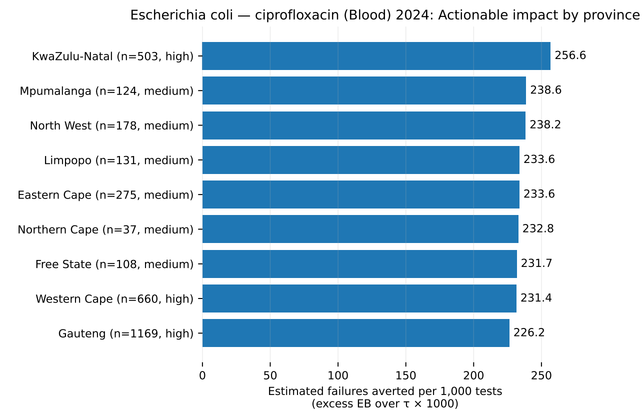

Figure 2: **Actionable impact (2024).** *E. coli*–ciprofloxacin (blood culture). Bars show empiric-treatment failures averted per 1,000 *tests* if guidance is updated, computed as $1000 \cdot \max(\hat{\theta} - \tau, 0)$. Confidence tiers reflect the pipeline's probability/stability logic; entries marked stable met the persistence criterion (two consecutive years at $\Pr(\theta > \tau) \geq 0.80$).

3.3 Top actionable items (ranked)

Table 1 lists the 2024 province entries ordered by impact per 1,000 tests, with EB point estimates, the threshold τ, absolute exceedance Δ (percentage points), confidence tier, and the composite stability flag.

3.4 Interpreting confidence and stability

All nine provinces are marked *stable* in 2024 (Table 1), meaning the exceedance criterion held for at least two consecutive years. Confidence tiers align with both statistical certainty and data support: large-n provinces (e.g., Gauteng $n = 1169$, Western Cape $n = 660$) are assigned *High* confidence; smaller-n provinces retain *Medium* confidence despite large Δ, reflecting conservative uncertainty handling. Notably, KwaZulu–Natal ($n = 503$) combines a large exceedance (25.7 pp) with High confidence, landing at the top of the actionable ranking.

3.5 Sample size, shrinkage, and prioritisation

The impact metric is expressed *per 1,000 tests*, so its value is driven by the *magnitude* of exceedance, not the number of tests in a given province. Nevertheless, n influences the *confidence* assigned to each call via the posterior and stability logic. EB shrinkage tempers volatility for smaller provinces (e.g., Northern Cape, $n = 37$), preventing spurious promotion while allowing persistent, large exceedances to surface with a Medium tier. Translating per-1,000 impacts into absolute expected failures averted requires multiplying by anticipated test volumes in each province.

Table 1: **Top actionable province entries (2024):** *E. coli*–ciprofloxacin (blood culture). Δ (pp) is EB minus threshold in percentage points. Impact/1,000 is $1000 \cdot \max(\hat{\theta} - \tau, 0)$.

Province	n	EB %	τ (%)	Δ (pp)	Impact/1,000	Confidence	Stability
KwaZulu–Natal	503	35.7	10	25.7	256.6	High	✓
Mpumalanga	124	33.9	10	23.9	238.6	Medium	✓
North West	178	33.8	10	23.8	238.2	Medium	✓
Limpopo	131	33.4	10	23.4	233.6	Medium	✓
Eastern Cape	275	33.4	10	23.4	233.6	Medium	✓
Northern Cape	37	33.3	10	23.3	232.8	Medium	✓
Free State	108	33.2	10	23.2	231.7	Medium	✓
Western Cape	660	33.1	10	23.1	231.4	High	✓
Gauteng	1169	32.6	10	22.6	226.2	High	✓

3.6 Cross-year consistency

Because the composite stability rule uses two consecutive years at $\Pr(\theta > \tau) \geq 0.80$, 2024 stability implies that exceedance was already present in the preceding year(s). Figure 1 provides the spatial baseline for 2023; Figure 2 then quantifies the 2024 actionable impacts. Together, they indicate sustained, above-threshold resistance across all provinces for this organism–drug–specimen, making guideline review system-wide rather than isolated.

4 Discussion

4.1 What this study shows

We introduced a closed-form, threshold-aligned decision rule that converts routine antibiograms into *stable, actionable* policy signals. By combining Empirical–Bayes (EB) shrinkage with the exceedance probability $\Pr(\theta > \tau)$ and a *composite stability rule*—promoting an alert if *either* two consecutive years satisfy $\Pr(\theta > \tau) \geq 0.80$ (*persistence*) *or* the **exceedance probability** ($\Pr(\theta > \tau)$) exhibits a positive three-year slope—the framework directly answers the stewardship question: *is true resistance in this province this year above the clinical threshold, and is that signal persistent rather than transient?* Applied to *E. coli*–ciprofloxacin (blood culture), 2023–2024, the method generated a compact, rank-ordered list of province-level actions prioritised by expected failures per 1,000 tests—evidence ready for scheduled guideline review [15].

4.2 How this improves decision quality

Traditional summaries (raw percentages, confidence intervals, rolling means) are unstable at small n and not aligned to decision thresholds [16]. Our approach improves on standard practice by:

1. **Calibration at small n.** EB pooling against national totals reduces variance for data-sparse provinces without suppressing sustained signals, shrinking noisy estimates toward a data-driven prior [25–27].
2. **Decision alignment.** $\Pr(\theta > \tau)$ is threshold-centric and interpretable as the probability that true resistance exceeds the clinical concern threshold—consistent with decision-analytic approaches in health policy [8].
3. **False-alarm control.** Requiring persistence *or* a positive three-year slope implements run-length and trend logic from statistical process control, trading slight detection delay for fewer premature alerts [13, 14].

Expressing priority as *excess failures per* 1,000 *tests* converts statistical differences into an interpretable *burden metric* that stewardship teams can triage [8].

4.3 Interpreting the 2023–2024 case study

In the current run, provinces exceeded the *learned* UTI threshold (τ) by substantial margins, satisfied the composite stability rule, and generally received High or Medium confidence tiers. This pattern signals a *system-wide misalignment* between empiric therapy and observed resistance for *E. coli* bloodstream infection, supporting a coordinated guideline update sequenced by expected impact per 1,000 tests and local testing volume [15].

4.4 Robustness and sensitivity

Because the pipeline is deterministic and closed-form (no MCMC), it is reproducible and auditable. Sensitivity analyses can vary τ (± 5 points), the exceedance cut-off ($P^* \in \{0.70, 0.80, 0.90\}$), the persistence window ($K \in \{2, 3\}$), and the slope window (2–4 years) to evaluate robustness [13, 25]. Future simulation studies could quantify false-alarm and missed-change rates under realistic n and drift scenarios to guide minimal sample sizes (n^*) for reliable provincial calls [28].

4.5 Limitations and threats to validity

The analysis draws on *public-sector* NHLS data; private-sector patterns may diverge [17]. Breakpoint updates and reporting shifts could alter apparent resistance. EB pooling may bias extreme provinces slightly toward the mean, a known trade-off in hierarchical models [25, 26]. Although the composite rule mitigates false positives, formal false-discovery control could be integrated. Threshold values (τ) should remain tied to clinical consequence and documented within stewardship protocols.

4.6 Portability and generalisation

The method requires only (n, r), a clinically meaningful τ, and location–year identifiers. It scales to other pathogens and settings with hierarchical structures (districts, facilities) and can run automatically on national dashboards at regular intervals [15, 24].

4.7 Policy implications

Stewardship committees can:

1. **Prioritise by burden** — rank pairs by excess failures per 1,000 tests to maximise reduction in empiric-failure risk.
2. **Tier by confidence** — implement high-confidence actions first; monitor medium-confidence provinces to improve certainty.
3. **Institutionalise cadence** — re-run analyses every 6–12 months with fixed thresholds for transparent, repeatable updates.

4.8 Future work

Planned extensions include (i) simulation-based power analysis to define n^*, (ii) validation against prescribing and consumption data, and (iii) forecast-aware variants estimating $\Pr(\theta_{t+1} > \tau)$ for proactive stewardship [24, 28].

4.9 Conclusions

A threshold-aligned, uncertainty-aware EB framework using exceedance probabilities and a composite persistence-or-slope rule delivers reproducible, decision-ready AMR signals. In *E. coli* bloodstream infections it prioritises a coordinated national empiric-therapy revision. Because the workflow is containerised and open, it supports transparent, auditable guideline governance across health systems [24].

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